

Steps followed in jupyter notebooks - CapstonePart1.ipynb and CapstonePart2.ipynb

- Data Preparation: Steps 1–5
- Data Modeling: Steps 6–11
- Business Insights: Steps 12–13

1. Data Loading and Exploration

- * Load the Dataset.
- * Check data types and missing values.
- * check first few rows.

2. Data Cleaning

- * Handle missing values.
- * Correct data types.
- * Remove duplicates.
- * Detect and optionally handle outliers.

3. Exploratory Data Analysis (EDA)

- * Univariate plots (histograms, boxplots).
- * Bivariate/multivariate plots (scatterplots, heatmaps).
- * Identify patterns and relationships.

4. Feature Engineering

- * Create new features (e.g., RFM, tenure, average satisfaction).
- * Encode categorical variables.
- * Scale/normalize features if needed.

5. Feature Selection

- * Correlation analysis.
- * Recursive Feature Elimination (optional).
- * Keep only the most useful features.

6. Data Splitting

- * Split data into Train, Validation, and Test sets (e.g., 60/20/20 or 70/15/15).

7. Baseline Model

- * Build a dummy baseline model (DummyClassifier/Regressor).
- * Train a simple model (Logistic Regression, Decision Tree, etc.).
- * evaluate using appropriate metrics (Accuracy/Precision/Recall/F1/ROC-AUC for classification, RMSE/MAE/ R^2 for regression).

8. Check Class Imbalance (if classification)

- * Apply resampling or adjust metrics if needed.

9. Prepare Data for Clustering

- * Standardize features (scaling).
- * Apply PCA (optional, for dimensionality reduction).

10. Customer Segmentation

Apply clustering algorithms:

- * K-Means
- * DBSCAN
- * Hierarchical Clustering

11. Validate clustering results:

- * Elbow Method
- * Silhouette Score

12. Analyze and Profile Segments

- * Define and label customer segments (e.g., "High Value Frequent Buyers", "Bargain Hunters").
- * Analyze demographics and purchase behavior per segment.
- * Suggest targeted marketing actions for each group.

13. Recommendation & Conclusion

- * How marketing can personalize offers based on segment profiles. *
- Predict churners based on low engagement.
- * Recommend targeted campaigns to maximize CLTV.